LAKSHIMA MADURANGI

SOFTWARE ENGINEER GRADUATE

CONTACT

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github.com/Lakshima409

PORTFOLIO

https://lakshima409.github.io/lakshima-portfolio/

SOFT SKILLS

- Analytical and problem-solving skills
- Quick learner and proactive attitude
- Team collaboration
- Independent working
- Attention to detail
- Written communication
- Time management
- Continuous learning

PROFILE

Software Engineering graduate (BEng Hons, London Metropolitan University) with a focus on machine learning and data analytics. Built PantryPal, an AI-powered recipe recommendation system using a hybrid approach (content-based and collaborative filtering, TF-IDF, cosine similarity), and carried out a quantitative big data research project in SPSS covering regression, correlation and reliability analysis. Skilled in Python, SQL, PyTorch, and Git, with a strong foundation in data preprocessing, model evaluation and full-stack development. Quick to learn new techniques, with a keen interest in computer vision and time-series analysis. Comfortable working independently or in a team, documenting experiments clearly, and checking results carefully.

EDUCATION

2025-2026

Bachelor of Engineering (Hons) in Software Engineering – Upper Second Class Honours

London Metropolitan University

Relevant Modules:
• Advanced Software Engineering
• Application Development
• Artificial Intelligence
• Project Analysis and Practice
• Final Year Research Project

2023 - 2024 September

Higher National Diploma (HND) in Computing
ESOFT Metro Campus

2023

Level 3 Diploma in Information Technology (DITEC)
ESOFT Metro Campus

2022

English Certificate Course
NIBM

Currently Studying

- Google Data Analytics Professional Certificate (Coursera)
- Google Advanced Data Analytics Certificate (Coursera)
- Cisco Certification Courses

TECHNICAL SKILLS

- Web Development:

HTML, CSS, React.js, Node.js, Express.js, Flask, ASP.NET MVC, Bootstrap 5

- APIs & Development:

REST APIs, Full-Stack Development, Authentication, Role-Based Access Control

- Databases:

MySQL, SQL Server, Database Design, Relational Databases

- AI & Data:

Machine Learning, PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, SciPy, Data Preprocessing, Model Evaluation, Data Analytics, Predictive Analytics, SPSS

- Programming:

Python, C#, JavaScript, PHP, SQL

- Software Engineering:

System Analysis, Requirements Engineering, Software Testing, Agile, UML, OOP, Design Patterns

- Digital & Other:

SEO, Microsoft Office, Research, Documentation, Project Management

- Tools & Environment

Git, GitHub, Jupyter Notebook, Google Colab, VS Code, PyCharm, Visual Studio, Microsoft Excel, Ms Project

- Data Visualization

Matplotlib

LANGUAGES

- Sinhala: Native
- English: Professional working proficiency

ACADEMIC PROJECTS

PantryPal – AI Recipe Recommendation & Meal Planning System | React.js, Node.js, Express.js, Python, Flask, scikit-learn, NumPy, MySQL, Docker

Final-Year Research Project Assessment: Proposal — B | Interim Report — B | Final Report — A

- Developed a full-stack AI-powered web application providing personalized recipe recommendations based on users' pantry items, dietary preferences, and previous interactions.
- Designed the system architecture, database structure, application workflow, and frontend/backend components using React.js, Node.js, Express.js, Flask, and MySQL.
- Implemented a hybrid recommendation approach combining content-based filtering, collaborative filtering, TF-IDF, cosine similarity, and machine-learning techniques to generate personalized recipe recommendations.
- Developed features including pantry management, dietary personalization, meal planning, user authentication, and automated shopping-list generation through REST API integration.
- Conducted software testing, system evaluation, and research analysis to assess functionality, recommendation performance, and overall system effectiveness.

AI Mental Health Chatbot | NLP, AI, Knowledge-Based Systems, Python, PyTorch, NLTK, NumPy, Flask, SQLite

- Collaborated on a conversational AI chatbot using natural language processing, knowledge-based reasoning, and context-aware response generation.
- Contributed to system analysis, architecture, implementation, knowledge-base design, testing, and technical documentation.
- Implemented conversation tracking and persistent user-data functionality as part of a team-based university project.

Big Data in Retail – Research Project | SPSS, Python, Data Analytics

- Investigated the impact of big data on retail marketing strategies, consumer behaviour, segmentation, and personalization.
- Analyzed quantitative research data using SPSS, including descriptive statistics, correlation, regression, and Cronbach's alpha reliability analysis.
- Evaluated predictive analytics and ethical considerations related to consumer data, privacy, and responsible data usage in retail.

eShift – Household Goods Transport Management System | C#, Windows Forms, SQL Server

- Developed a desktop transport management system to automate booking, vehicle and driver management, route planning, delivery tracking, and reporting.
- Implemented role-based authentication, secure password handling, database normalization, and layered architecture.
- Applied object-oriented programming, design patterns, unit testing, and business logic implementation to develop a maintainable application.

ACADEMIC PROJECTS

Sampath Food City Management System | C#, OOP, Design Patterns, UML

- **D**eveloped a scalable retail management system using object-oriented programming and software engineering principles.
- Applied SOLID principles and Factory and Strategy design patterns to create maintainable, flexible, and reusable system components.
- Designed UML models and applied software architecture principles to support modularity, scalability, and maintainability.

Luxora Hotel Suite – Property Management System | PHP, MySQL, Bootstrap 5, REST APIs

- Collaborated in a four-person Agile team to develop a full-stack hotel reservation and property management system.
- Developed billing, check-in/check-out, dynamic invoicing, split payments, tax calculations, late-checkout penalties, and reservation handling.
- Implemented role-based access control, audit logging, database normalization, and responsive web interfaces.

Bus Seat Reservation System – Data Structures & Algorithms | Java, Data Structures, Algorithms

- Developed a Java-based bus seat reservation system for managing passenger bookings and seat allocation.
- Implemented stacks, queues, linked lists, and graphs to efficiently manage dynamic reservation and passenger data.
- Applied graph-based shortest-path algorithms and algorithmic problem-solving techniques to support efficient route management.

Photographer's Website - HTML, CSS, JavaScript, PHP, MySQL, SEO

- Designed and developed a responsive photographer portfolio website using HTML, CSS, JavaScript, PHP and MySQL, with a focus on usability and responsive presentation.
- Implemented CMS-based content management and structured the website to support efficient content updates and maintenance.
- Applied basic SEO optimization techniques to improve website structure, content visibility and search-engine accessibility.

Event Ticket Online Web Application | ASP.NET MVC, C#, SQL, Full-Stack Development

- Developed a scalable event-ticketing platform supporting event management, venue configuration, ticket booking, seat selection, QR-code e-tickets, and customer booking history.
- Implemented role-based authentication, payment processing, notifications, loyalty features, promotional codes, and analytics functionality.
- Applied clean architecture and modular design principles to support maintainability and scalability.